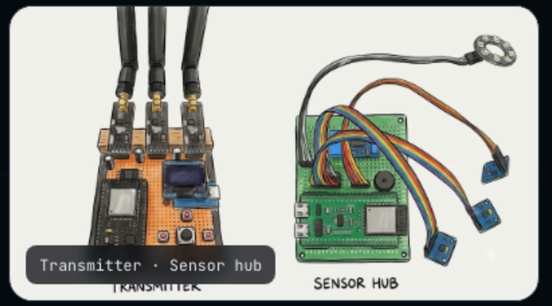
GEMINI TRACK · TIGERDATA TRACK · LIVE HARDWARE

# Cyber Chaukidaar

Ground-sensing sentries that tell a person from an animal, a tunnel from a footstep, and a drill from a door slam. Then explain it in plain words on your phone.




Node N1 online · Panel fw 2.2.0 · Gemini + TigerData live

THE PROBLEM

# Cameras and guards watch the surface. The threat arrives through the ground.




NUCLEAR EXCLUSION ZONES

## A guard in the hazard

A person stands a shift in a radiation area, mostly to tell wildlife from a trespasser.

BORDERS

## Tunnels under the fence

Digging happens beneath cameras and patrols. Ground vibration is the earliest evidence.

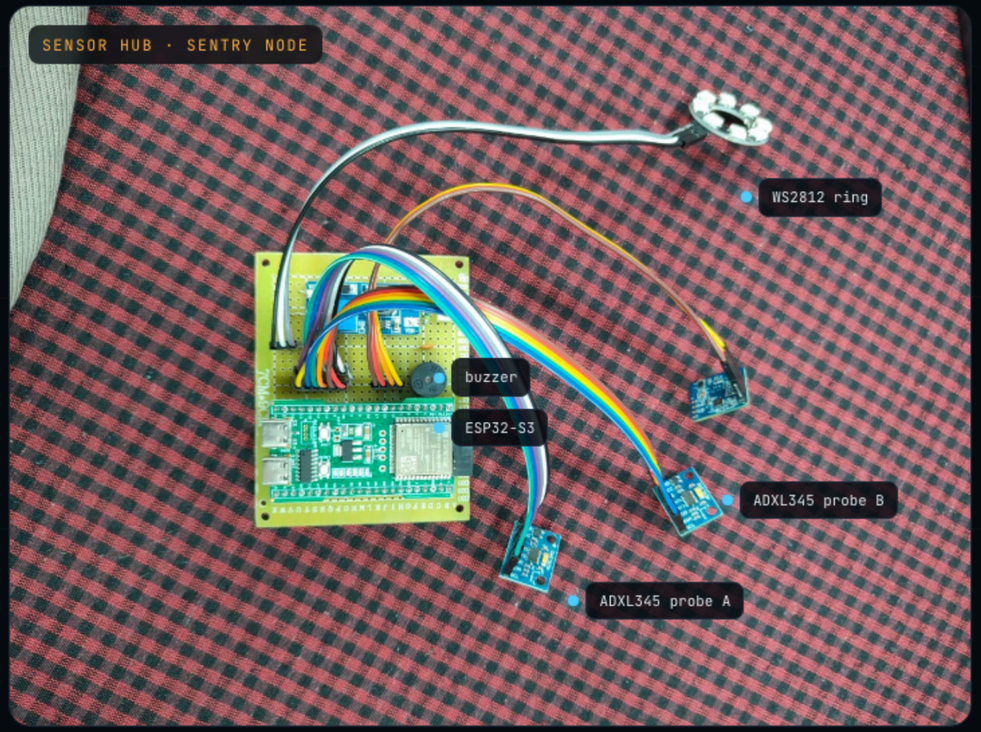
JEWELRY STORES

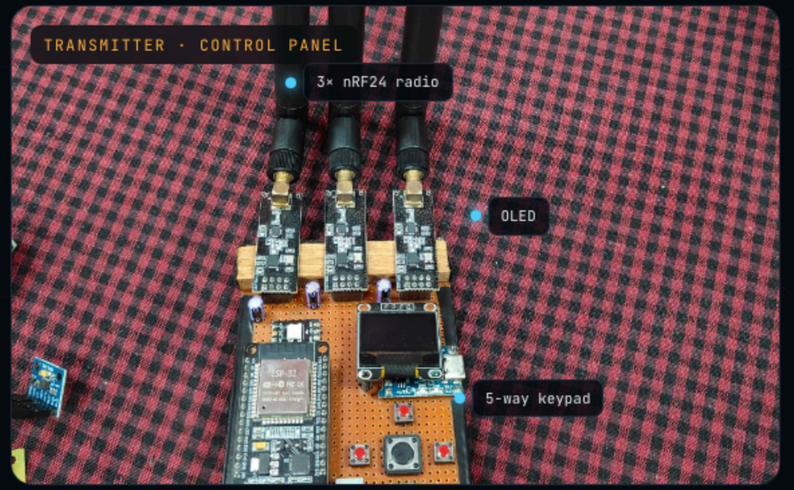
## In through the wall

After hours the entry point is a shared wall or the vault, where no camera is pointed.

THE HARDWARE, BUILT AND RUNNING

# Two boards we soldered this week




**Node** ESP32-S3, 2× ADXL345 (SPI), LD2420 radar, ring, buzzer, 18650 or USB

**Panel** ESP32, 3× SSD1306, 5 keys, status pixel, nRF24 modules

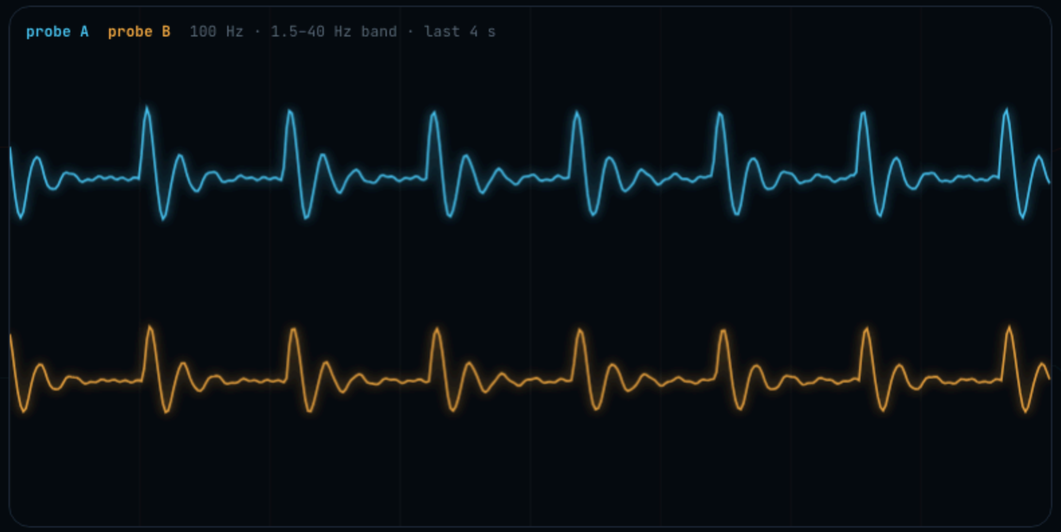
**Server** Node.js on a laptop or Raspberry Pi

**Links** WiFi now; BLE and LoRa drivers in the firmware

WHAT THE GROUND SOUNDS LIKE

# Four threats, four signatures. Pick one.

Person walking | Small animal | Digging (tunnel) | Drill on a wall






| IMPACT GAP | DOMINANT | CREST |
|---|---|---|
| 550 ms | 8 Hz | 4.1 |

## Unauthorized person in hazard zone

Slow, heavy, low-pitched steps; 3 of 4 gait cues; radar confirms a surface target.

CRITICAL · siren

HOW IT WORKS

# From a footstep to a decision in about a second

1 · NODE

### Feel and classify

Band-pass, STA/LTA trigger, 15 features, random forest on the chip, probe A/B timing.

2 · FUSION

### Re-check and fuse

The server re-runs the forest, fuses seismic and radar with an explainable logistic regression.

3 · MISSION

### Apply the use case

Gait vote, tunnelling, drilling and impact rules. Every alert carries its reason in words.

4 · RESPOND

### Act in the right way

Siren or silent alarm, phone app, OLED panel, WebSocket dashboard.

G **Gemini**
Explains every real alert for the active mission, in the alert path.

T **TigerData**
Stores every reading and alert with a SHA-256 evidence hash.

THREE MISSIONS ON ONE DESIGN

# Same hardware, different rules, different response

| MISSION | HOW IT DECIDES | RESPONSE |
|---|---|---|
| **Nuclear zone** | Gait score from 4 bipedal cues, majority vote over the last 5 step windows. Animals step faster, lighter, higher-pitched. | Siren from MEDIUM to scare wildlife off. Any person is CRITICAL. |
| **Border tunnel** | Rhythmic, low-frequency impacts from a source that does not move, nothing on the radar, sustained 4+ windows. | Silent alarm, so diggers are not warned. |
| **Jewelry store** | Continuous high-frequency vibration over 4× ambient for 3 windows; sharp impacts; a target in the showroom after hours. | "Drilling on wall/vault" CRITICAL, 15 s siren. |
| **Perimeter** | People and vehicles crossing, zones, loitering, tamper. | Siren from HIGH. |

Every alert carries its reason, for example "Rhythmic impacts every ~630 ms, source stationary, nothing on the surface radar".

GEMINI TRACK

# Gemini turns sensor numbers into an operator's answer

```
alert #2041 · N1 · mission: Border & Tunnel Watch

Rhythmic low-frequency impacts every ~630 ms from a source that
has not moved, with nothing on the surface radar, now sustained
for 135 windows. This matches digging rather than a walker.
Treat as possible tunnelling near N1. Silent alarm kept: do not
sound sirens. Dispatch a patrol to the north fence and watch N2
for the same rhythm.

> how many alerts at the north gate tonight?

SELECT count(*) FROM alerts WHERE node_id = 'N1' AND t > now() -
interval '12 hours'; → 3
```

### ALERT NARRATION

Every real alert explained for the active mission: what, where, how sure, what to do. Streamed live over WebSocket.

### SITUATION REPORT

One click summarises the site: nodes, open alerts, recent activity.

### ASK THE DATA

Plain-English questions become one guarded, read-only SQL query on TigerData.

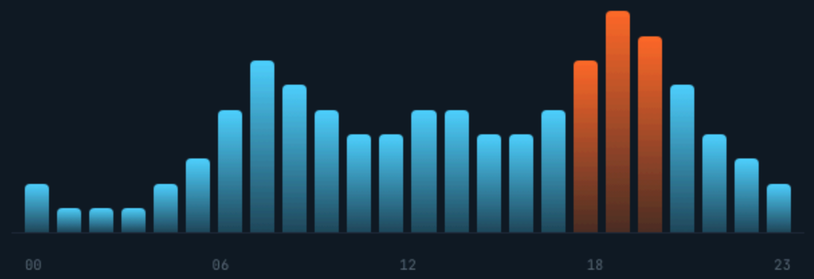
### ENGINEERING

gemini-flash-lite-latest · 12 s timeout · quota cooldown · template fallback, so an alarm never waits on the AI.

TIGERDATA TRACK

# TigerData is the memory and the evidence locker

ACTIVITY PER HOUR · FROM THE TELEMETRY_1S CONTINUOUS AGGREGATE (ILLUSTRATIVE SHAPE)




| | |
|---|---|
| **hypertables** | every telemetry packet and every alert, partitioned by time |
| **telemetry_1s** | continuous aggregate refreshed every 10 s; charts never scan raw rows |
| **compression** | chunks older than 1 day, segmented by node |
| **retention** | raw rows dropped after 30 days, alerts kept |
| **sha-256** | evidence hash stored with each alert, mission and label |
| **read-only role** | the only door Gemini's SQL can use |

LIFE OF A TELEMETRY CHUNK · ONE CELL PER DAY



day 0 · raw rows, 1 s aggregate feeds the charts

day 1 → 29 · compressed, segmented by node

day 30 · raw dropped, alerts and hashes kept

BUSINESS MODEL

# Hardware once. AI review every month.

DAY 0 · ONE-TIME

## Hardware + installation

| | |
|---|---|
| Sentry node (probes, radar, ring, siren) | ₹7,999 |
| Control panel (OLED, keypad) | ₹4,999 |
| Site survey, spikes, coupling, calibration | ₹6,000 / site |

Example pricing. Approximate node BOM ≈ ₹2,500 (ESP32-S3, 2× ADXL345, LD2420, ring, buzzer, cell, box), so hardware pays for itself and funds installation.

MONTH 1+ · SAAS

## AI review subscription

| | |
|---|---|
| **Watch** · alerts, phone app, 30-day history | ₹999 / site / mo |
| **Review** · Gemini narration, weekly AI incident review, ask-the-data | ₹2,999 / site / mo |
| **Enterprise** · silent-alarm integrations, long retention, SLA | custom |

Recurring revenue scales with sites, not with staff. Gemini flash-lite and TigerData compression keep the cost per site in rupees, not thousands.

install → alerts + AI reviews → labelled field data → better models → fewer false alarms → more sites

THE TEAM

# Order of Phoenix


IS

**Ishpreet Singh**

Order of Phoenix

PS

**Prabhmehar Singh Sahni**

Order of Phoenix

VC

**Vansh Chawla**

Order of Phoenix

AM

**Avirath Magoo**

Order of Phoenix


`firmware · C++ / PlatformIO` `server · Node.js` `dashboard · React PWA` `ML · Python` `hardware · perfboard, soldered by hand`